The Failure Point of Systems of Linearized Differential Equations

How far from equilibrium does the linearization of a two-variable energy-balance climate model remain a valid predictor of the system's behaviour?

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1 Introduction

2 Modelling

With the aim of understanding how the number of rounds affect the orderedness of the cards in play, I will be modelling the following relationship:

x-axis: the number of rounds played (independent variable)
y-axis: the cards flipped from the draw pile (dependent variable)

I am using the variable of cards flipped from the draw pile because this is a key indicator that isolates the orderedness of only the cards in play rather than assessing the orderedness of the whole deck, reducing the scope and therefore increasing the validity.